Hemanth Sai Korapala

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AI/ML professional with 4 years of expertise in developing and fine-tuning LLMs (BERT, GPT) and multimodal AI models. Skilled in NLP tasks such as sentiment analysis, language modeling, and machine translation. Proven experience with Generative AI models, cloud platforms, and big data tools to build scalable AI systems. Strong communicator with a passion for driving innovation and solving real-world challenges in AI/ML.

TECHNICAL SKILLS

Programming languages: Python, MATLAB
ML Architectures: Transformers, LLM, Multi-modal LLMs, Neural Networks
Frameworks: Langchain, Llama Index, CrewAI
LLMs: BERT, GPT, Mistral, Google Palm
Databases: SQL, Pinecone, Chroma
Containerization tools: Docker, Kubernetes
CI/CD tools: Gtihub Actions, Jenkins
Data Tools: Apache Spark, Hadoop
Cloud: AWS services (S3, EC2, Bedrock, and Sagemaker), Microsoft Azure, GCP, Snowflake
Libraries: Hugging face, TensorFlow, PyTorch, NLTK, keras, Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV
NLP Tasks: Named Entity Recognition (NER), Machine Translation, Sentiment Analysis, Language Modeling
Machine Learning algorithms: Regression and Classification Algorithms, Clustering, Support Vector Machines, Naive Bayes

WORK EXPERIENCE

Machine Learning Engineer — *April 2024 – Present*
Omdena

- Developed "Mindguardian," a mental health counseling chatbot leveraging state-of-the-art AI techniques, including advanced prompt engineering, Sentence Transformer, Google Gemini, and Chroma DB.
- Applied MLOps principles for streamlining the development, deployment, and maintenance of machine learning models.
- Containerized the chatbot using Docker to ensure portability and consistency across environments.
- Implemented automated CI/CD pipelines using GitHub Actions for seamless builds, testing, and deployments to Amazon ECR and AWS SageMaker.
- Deployed scalable and secure applications on AWS, utilizing services such as EC2, ECR, and IAM with security best practices.
- Designed and orchestrated containerized workloads on Kubernetes, optimizing resource allocation and ensuring fault-tolerant deployments.
- Leveraged Kubernetes features such as ConfigMaps, Secrets, and Ingress Controllers to manage configurations and expose applications efficiently.
- Implemented custom embedding functions and efficient batch-processing pipelines for Chroma DB, enhancing the scalability of text data management systems.
- Conducted advanced sentiment analysis on El Salvador's political landscape using social media data and ensemble techniques (Random Forest, XGBoost, GBM).
- Fine-tuned BETO (Spanish BERT) using PyTorch for improved accuracy, leveraging PEFT techniques like LoRA and Q-LoRA for model optimization.
- Developed Generative AI models for synthetic data generation, improving model robustness and handling underrepresented data scenarios.
- Collaborated with MLOps teams to integrate machine learning solutions into production environments and streamline end-to-end workflows.
- Designed robust and scalable software architectures, implementing text representation techniques like n-grams, bag-of-words, and sentiment analysis for NLP tasks.
- Wrote efficient, testable, and maintainable Python code to ensure high-quality implementations.

Machine Learning Engineer *May 2024 – Present*
Staffna

- Developing an audio-to-audio chatbot to automate and streamline repetitive tasks in industrial environments, integrating Speech-to-Text (Whisper), text processing (Mistral 7B), and Text-to-Speech (TTS) models.
- Created a Retrieval-Augmented Generation (RAG) system utilizing Mistral 7B for text-to-text tasks using Llama Index platform by incorporating various retrievers and retrieval technique called Hypothetical Document Embeddings (HyDE).
- Integrating Speech recognition for Speech-to-Text (STT) processing and implementing Text-to-Speech (TTS) models to generate audio responses.
- Managed cloud resources efficiently to optimize performance and cost.
- Collaborated with cross-functional teams to align machine learning solutions with business objectives.

Research Intern *January 2023 – December 2023*
Saint Louis University

- Worked on research to implement Irish Celtic mutation as a style transfer task in NLP, preserving content while infusing unique linguistic characteristics.
- Worked on fine-tuning language models (LLMs) like GPT, BERT, and other transformers.
- Leverage TensorFlow, Keras, and NLTK to fine-tune a large language model, achieving remarkable proficiency in text generation with cultural authenticity.
- Evaluated the model performance using metrics such as BLEU score and ROUGE score.
- Contributed to the advancement of NLP frameworks while promoting linguistic diversity and cultural heritage preservation.

Data Scientist *October2018 - November 2021*
Client: HSBC

- Worked on AML data and developed logistic regression models using Python to detect suspicious accounts (SAR), improving fraud detection accuracy.
- Extracted data from the AWS S3 service and efficiently processed it in a Python environment, ensuring seamless data integration. Conducted thorough feature and variable analysis to identify key predictors for the developed models.
- Evaluated model effectiveness using Precision, Recall and F1 score ensuring reliable predictions, and minimizing false positives and false negatives.
- Created comprehensive model documentation, presenting complex concepts in an easily understandable manner for senior management and non-technical stakeholders. Worked closely with the model risk management team, actively identifying and mitigating risks through the application of robust risk models.
- Utilized Python libraries and packages such as pandas, NumPy, scikit-learn, and matplotlib for data manipulation, analysis, visualization, and model development.

EDUCATION

M.S. Artificial Intelligence (Computer Science), Saint Louis University, GPA 3.6/4.0 *December 2023*
***Relevant Coursework*:** Machine Learning, Natural language processing, Computer vision, Big Data, Deep learning, Artificial intelligence, Software development, Agile methodologies

B.Tech. Electronics and Communications, Vignan's University, GPA 7.8/10 *May 2018*
***Relevant Coursework*:** Image Processing, Digital Signal processing, Mathematics